NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

QUIZ 2: L5-L6

INSTRUCTIONS TO STUDENTS

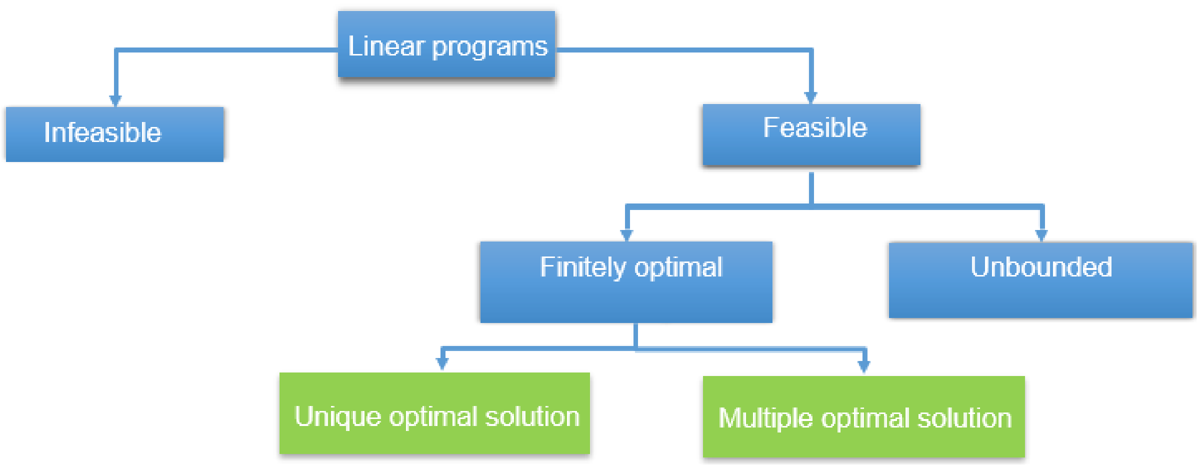
1. This assessment consists of four types of questions:
True/False Questions, Multiple Choice Questions (only one answer is correct), Multiple Response Question (allow for multiple correct answers), and Fill in the Blank Question.
2. There are a total of 2 True/False Questions (3 marks each), 2 Multiple Choice Questions (5 marks each), 6 Multiple Response Questions (8 marks each), and 2 Fill in the Blank Questions (12 marks each), for a total of 88 marks. For MRQ (Multiple-Response Questions), marks will be deducted for any incorrect options selected — but your score for each MRQ will never go below zero.
4. This is an open-book assessment. You may use printed or soft copy materials (e.g., lecture slides, textbooks, notes). However, you are not allowed to search the internet or use any LLM-powered tools (e.g., ChatGPT or other language models).
5. Other devices (e.g., iPads and phones) are not allowed. Please note that graphing calculators are not permitted. A basic calculator is provided for numerical calculations, accessible under "Tool Kit" at the top right of the exam interface.
6. Time allowed: 30 minutes.

Question 1

True or false: If a linear program is both feasible and bounded, then it must have at least one optimal solution.

(a) True
(b) False

Ans: (a). Please refer to the figure below.



Question 2

True or false: In a linear program with two variables, if the slope of the level set of the objective function is identical to the slope of one of the constraints, we will always have multiple optimal solutions.

(a) True
(b) False

Ans: (b). There are different cases:

Case 1: Optimal edge coincidence (leads to multiple solutions)

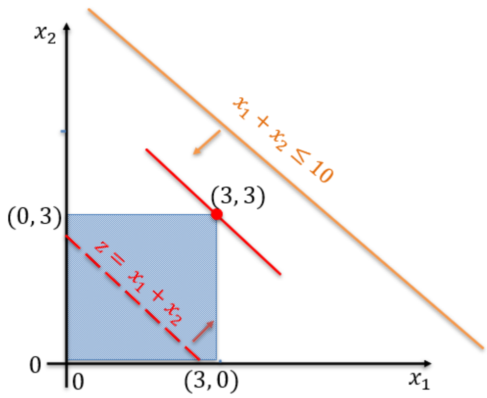
Minimize $z = x_1 + 2x_2$
subject to:

$$x_1 + 2x_2 \geq 6$$
$$2x_1 + x_2 \geq 6$$
$$x_1, x_2 \geq 0$$

Case 2: Parallel but non-active constraint (unique solution).
The objective function's slope is identical to a constraint, but that specific constraint is not binding (active) at the optimal solution. The optimal solution might be at a vertex determined by the intersection of *other* constraints. In this case, even though a parallel constraint exists, it does not influence the specific optimal vertex, and you would have a unique optimal solution.

Maximize $z = x_1 + x_2$
subject to:

$$x_1 + x_2 \leq 10$$
$$x_1, x_2 \leq 3$$
$$x_1, x_2 \geq 0$$

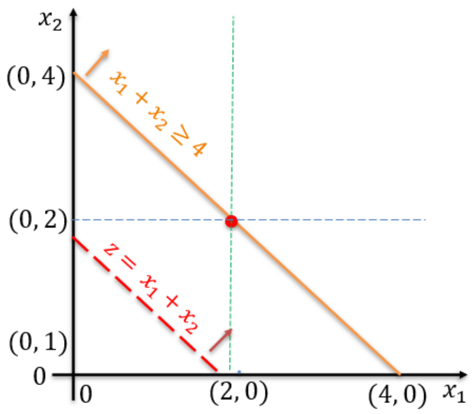


Case 3: Parallel, the constraint is active, but optimum is a single vertex (unique solution).
The objective function's slope is identical to a constraint, and that specific constraint is binding (active). However, the feasible region is only a single vertex.

Maximize $z = x_1 + x_2$
subject to:

$$x_1 + x_2 \geq 4$$
$$x_1 \leq 2$$

$$x_2 \leq 2$$
$$x_1, x_2 \geq 0$$



Question 3

In a linear programming (LP) problem, which of the following must always be true? Select all that apply.

(a) The feasible region is always bounded if the problem has an optimal solution.

(b) The objective function and all constraints must be linear.

(c) If a linear program has an optimal solution, then there exists at least one optimal solution at a vertex (or along an edge connecting optimal vertices) of the feasible region.

(d) The constraints can be written as strict inequalities ($<$, $>$) or non-strict inequalities ($\leq$, $\geq$).

Ans: (b) (c)

Question 4

Select all options that are linear programs. (More than one answer may be correct.)

(a) Maximize: $12A + 6B$

s.t.: $10A + 5B \leq 1100$

$600A + 400B \leq 420000$

$A, B \geq 0$

(b) Minimize: $800A + 300B$

s.t.: $10A + 5B \leq 1100$

$8A + 4B > 15$

$A, B \geq 0$

(c) Maximize: $2\sin(x) + 5\cos(y)$

s.t.: $11x + 7y \leq 12$

$x + y \geq 4$

$x,\ y \leq 10$

(d) Maximize: $x + y$

s.t.: $x^2 + y^2 \leq 11$

$$2x + 3y \le 20$$
$$x,\ y \ge 0$$

(e) Minimize $3x + 2y$

s.t.: $3x + 2y \le 15$

$4x + 3y \le 20$

$x,\ y \ge 0$

Ans: (a) (e). In a linear program, both the objective function and the constraints must be linear (affine). The constraints can be equalities (=) or non-strict inequalities (≤, ≥), but strict inequalities (<, >) are not allowed.

Question 5 (allow normal calculators)

Consider the following linear program (LP), select all the options that are correct. (More than one answer may be correct.)

Maximize: $z = 3x_1 + 4x_2$

Subject to:

$$x_1 + x_2 \le 4$$
$$x_1, x_2 \ge 0$$

(a) The maximum value of the objective function is 16.

(b) If we remove the constraint $x_1, x_2 \ge 0$, it is still an LP problem.

(c) If we remove the constraint $x_1 + x_2 \le 4$, it is still a feasible LP problem.

(d) If the constraints were changed to: $x_1 + x_2 < 4$, it is still a linear program.

(e) If we changed the problem's objective function to: Minimize $z = -3x_1 - 4x_2$, then the optimal solution for (x_1, x_2) remains unchanged.

(f) $(-1,\ 4)$ is a feasible but not the optimal solution for the above LP problem.

Ans: (a) (b) (c) (e). You can use either graphical method or Python program to solve this LP. Option (e) is correct because multiplying the objective function by −1 simply flips it from a maximization to a minimization (or vice versa), without changing the feasible region or the location of the optimal (x_1, x_2). The optimal solution stays the same, only the objective value's sign changes.

Question 6 Tournament Transportation Optimization

Your student club is organizing a massive inter-university gaming tournament with 400 participants. You need to arrange transportation from campus to the esports arena using two types of awesome vehicles:

- Party Bus:
  - Capacity: 50 players
  - Cost: $800 per bus
  - Requires 1 driver
  - Max 10 available
  - *Comes with RGB lighting, snack stations, and a built-in sound system.*
- Gamer Van:
  - Capacity: 40 players

- Cost: \$600 per van
- Requires 1 driver
- Max 8 available (they're popular!)
- *Equipped with individual charging ports, headphone sanitizers, and premium seating.*

Constraints you face:

1. Driver Shortage: You will use your own drivers from your club. Only 9 club members have driver licenses, and there is no extra budget to hire external drivers.
2. Budget Cap: The total rental cost cannot exceed \$6,000 (we know how club funds work!).
3. Non-Negotiable: You must transport all 400 participants.
4. Vehicle Mix Policy: The club president decrees that at least 40% of the vehicles must be Gamer Vans (to ensure enough charging ports).

Your mission is to minimize the total rental cost so you can afford the best prizes for the winners.

Let decision variables:

- B = number of Party Buses used
- V = number of Gamer Vans used

Which of the following is the correct formulation of this Linear Programming problem?
(Min is an abbreviation for Minimize)

(a) Min $800B + 600V$
s.t.:

$$
\begin{gathered}
B + V \leq 9 \\
50B + 40V \geq 400 \\
800B + 600V \leq 6000 \\
3V \geq 2B \\
B, V \geq 0 \\
B, V \in \mathbb{Z}
\end{gathered}
$$

(b) Min $800B + 600V$
s.t.:

$$
\begin{gathered}
B + V \leq 9 \\
50B + 40V = 400 \\
800B + 600V \leq 6000 \\
V \geq 0.4(B + V) \\
B, V \geq 0 \\
B, V \in \mathbb{Z}
\end{gathered}
$$

(c) Min $B + V$
s.t.:

$$
\begin{gathered}
B + V \leq 9 \\
50B + 40V = 400 \\
B \leq 10 \\
V \leq 8 \\
800B + 600V \leq 6000 \\
V \geq 0.4(B + V) \\
B, V \geq 0 \\
B, V \in \mathbb{Z}
\end{gathered}
$$

(d) Min $800B + 600V$
s.t.:

$$\begin{aligned} B + V &\leq 9 \\ 50B + 40V &\geq 400 \\ B &\leq 10 \\ V &\leq 8 \\ 800B + 600V &\leq 6000 \\ 3V &\geq 2B \\ B, V &\geq 0 \\ B, V &\in \mathbb{Z} \end{aligned}$$

(e) Min $B + V$
s.t.:

$$\begin{aligned} B + V &\leq 9 \\ 50B + 40V &\geq 400 \\ B &\leq 10 \\ V &\leq 8 \\ 800B + 600V &\leq 6000 \\ V &\geq 0.4(B+V) \\ B, V &\geq 0 \\ B, V &\in \mathbb{Z} \end{aligned}$$

Ans: (d).

Question 7

The given linear program is:

Maximize $z = 3x_1 + 2x_2$
subject to:

$$\begin{aligned} x_1 + x_2 &\leq 5 \\ 2x_1 + x_2 &\geq 12 \\ -x_1 + 2x_2 &\leq 4 \\ x_1 - x_2 &\leq 3 \\ x_1, x_2 &\geq 0 \end{aligned}$$

(a) Bounded and feasible
(b) Feasible but not bounded
(c) Bounded but not feasible
(d) Neither bounded nor feasible

Ans: (d). You can use either graphical method or Python program to solve this LP. Refer to Quiz2-Q7.py for sample implementation.

Question 8

Which of the following functions are convex? Select all that apply.

(a) $f(x) = x^2 - 4x + 10, x \in \mathcal{R}$
(b) $f(x) = |x - 3|, \ x \in \mathcal{R}$
(c) $f(x) = -x^2, x \in \mathcal{R}$

(d) $f(x) = \sqrt{x},\ x \in (0,\ +\infty)$

Ans: (a) (b)

Question 9

Which of the following statements about convex functions are true? Select all that apply.

(a) The sum *af(x)* + *bg(x)* is convex if functions *f(x)*, *g(x)* are convex and the coefficients *a* and *b* are positive (i.e., $a > 0$, $b > 0$).

(b) If *g(·)* is convex, then the function *f(x)* = *g(Ax + b),* which is a composition of *g(·)* with an affine transformation in the input space, is also convex.

(c) A convex function must be linear.

(d) For convex functions, any local minimum is also a global minimum.

(e) The product of two convex functions is always convex.

(f) Linear/affine functions are special cases of convex functions

Ans: (a) (b) (d) (f). Please refer to the properties of convex functions in the lecture notes. For convex optimization problems (such as linear and logistic regression), these properties ensure a single global minimum, making algorithms like gradient descent reliable and efficient by avoiding local minima.

Question 10

Which of the following statements about gradient descent are correct? Select all that apply.

(a) Gradient descent updates the function's variable by moving in the opposite direction of the gradient — that is, in the direction of steepest decrease of the function.

(b) For an unconstraint convex optimization problem with a differentiable objective function, gradient descent will converge to the global minimum if the learning rate is chosen appropriately.

(c) If the learning rate is too large, gradient descent might diverge or oscillate instead of converging.

(d) For non-convex functions, gradient descent may get stuck in local minima, but under certain conditions it can find reasonably good solutions.

(e) Gradient descent can work for any differentiable function, not just convex ones.

Ans: (a) (b) (c) (d) (e). *Reminder: This question focuses on gradient descent, a key concept in the Optimization topic—make sure you understand how it works and its main characteristics.*

Question 11: Weekly stock trading optimization (allow normal calculators)

A trading firm manages five different stocks this week. For each stock, the firm can choose to buy (positive units), sell (negative units), or make no trade (zero units). The firm currently holds a sufficient number of units of all five stocks to cover any potential sales allowed by company policy.

Note: Selling a stock is represented by negative values of x_i. For example, $x_2 = -3$ means selling 3 units of Stock 2

Each stock contributes differently to the weekly benefit:

Stock	**Variable**	**Contribution per unit (net profit)**	**Notes**
Stock 1	x_1	+$2	Buying gains $2 per unit
Stock 2	x_2	−$3	Buying loses $3 per unit; Selling gains $3 per unit
Stock 3	x_3	+$1	Buying gains $1 per unit
Stock 4	x_4	$0	Neutral — no gain or loss
Stock 5	x_5	$0	Neutral — no gain or loss

The firm faces the following constraints due to company policies and client agreements:

$$x_1 - x_2 + x_3 \leq 5$$
$$x_1 - x_2 + 4x_3 \leq 7$$
$$x_1 + 2x_2 - x_3 + x_4 \leq 14$$
$$x_3 - x_4 + x_5 \leq 7$$

Additional rules based on market policies:

$$-15 \leq x_i \leq 15 \; for\ all\ i$$

x_i must be integers.

Objective:

The trading firm hopes to maximize its total weekly net profit ($):

$$\text{Maximize} \quad 2x_1 - 3x_2 + x_3$$

Your task:

(a) Analyze the problem status of this integer linear programming problem.

(b) Solve this integer linear programming problem and report the maximum weekly net profit. *Round or report your answer as an integer.*

For part (a), select the correct choice from the dropdown menu.

For part (b), which is a fill-in-the-blank, your answer must contain only digits (0-9) and an optional minus sign (−) at the beginning.

Do not include any letters, symbols (such as $), spaces, or commas in your answer for part (b).

(a) The problem status:

i. The problem is feasible and bounded.

ii. The problem is feasible but unbounded.

iii. The problem is infeasible.

iv. The problem is bounded for the integer linear program, but unbounded for its LP relaxation.

(b) Maximum weekly net profit (in $): ______________

Ans: (a): The problem is feasible and bounded. (b): 40, refer to Quiz2-Q11.py for sample implementation.

There are multiple answers leading to the same objective value as below.

x1=15, x2=-5, x3=-5, x4=-15, x5=15, Profit=40

x1=5, x2=-15, x3=-15, x4=-15, x5=-15, Profit=40

When multiple optimal solutions exist that yield the same objective value, optimization solvers (accessed through libraries such as CVXPY or PuLP) typically return a single solution, determined by the solver's internal algorithm or numerical tolerances.

Question 12 (allow normal calculators)

Suppose we are minimizing $f(x) = (x + 2)^2 + 7$ with respect to x. We initialize $x = 4$, and perform gradient descent with learning rate $\eta = 0.01$.

Calculate the updated values of x after the first and second iterations. Round your answers to 2 decimal places.

Gentle reminder: For automated grading to work correctly:

- For each box, enter only a number.
- The number can include a negative sign (-), digits (0-9), and a decimal point.
- Do not include any letters, spaces, commas, or other characters.

Value of x after the first iteration: ______________

Value of x after the second iteration: ___________

Ans: The updated value of x after the second iteration is **3.88**. (acceptable range: 3.87-3.89)

The updated value of x after the second iteration is **3.76**. (acceptable range: 3.75-3.77)

To solve this, first compute the derivative of $f(x)$, $f'(x) = 2(x + 2)$. Then, apply the gradient descent update rule $\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k - \eta \nabla_x f(\mathbf{x}_k)$ iteratively: use $x = 4$ for the first iteration, calculate the updated x, and then use that value for the second iteration. Round each result to two decimal places as required. You can also use Python program to solve this question.